

SQL:

Tables - Stores the data in rows and columns

primary key - where it creates a unique identifier for each record

constraints - Rules applied to ensure the data is valid or not

There are a few command categories that sql follows when we use them

DDL - Data definition language

- where it deals with how we structure the data
the common cmds are
 - create - helps to create databases and its objects
 - Drop - helps to delete objects from the database
 - Alter - helps to alter the db structure
 - Truncate - helps to remove everything

DQL - Data query language

- where it helps to retrieve/fetch the data
 - Select - helps to retrieve the data from the database
 - From - helps to retrieve from which table
 - where - helps to filter the rows before grouping
 - Group by - helps to group the rows that have the same values
 - Having - helps to filter the rows on basis of group by
 - Distinct - removes duplicate rows from the query result only, the table data stays unchanged
 - order by - helps to sort the result rows of a query in asc/desc

DML - Data manipulation language

- where it helps to manipulate the data
 - insert - helps to insert the elements
 - update - helps to update the existing elements
 - delete - helps to remove the records from the db

DCL - Data Control language

- where it deals with the access control of the data
 - GRANT - assigns privileges to a certain user
 - REVOKE - removes the privileges from the account

TCL - Transaction control language

- where it allows us to run set of tasks into a single execution unit!
 - Begin Transaction - starts a new set of tasks
 - Commit - save the changes that happened during transactions
 - Rollback - undoes all the changes during transactions
 - savepoint - create save points within the current transaction

Constraints and types i used while creating tables

- Foreign key - connects a column of one table to the primary key of another table
- On delete cascade - when the parent row is deleted, child rows get deleted automatically
- Not null - the column must always have a value
- Default - fills a value by itself when we don't pass one (like current_date)
- Serial / Bigserial - auto generates id numbers for every new record
- Timestamptz - stores date-time with timezone so one instant means the same world
- Index - makes filtering fast on columns we search a lot (like machine_id + time)

Extra commands i used in the api projects

- Returning - gives back the row we just inserted/updated, no second select needed
- On conflict do nothing - skips the row if it already exists, duplicate safe insert
- On conflict do update - upsert, updates the existing row instead of throwing an error
- Limit / Offset - fetch only few rows and skip rows for pagination
- Min / Max / Avg - aggregate functions, used in the statistics endpoint
- Execute_values - batch insert 5000 rows in one shot for the big imports
- Rowcount - tells whether the update/delete actually changed any row
- Autocommit - needed for create database, postgres will not allow it inside a transaction

Not used yet - next to learn

- Joins - combining two tables in one query (so far my foreign key only linked the two tables)
- Group by / Having - learned in notes, not used in real code yet
- Window functions - over, partition by, row_number - not used yet
- Explain analyze - to see how postgres actually runs a query

Table partitioning - what i learned in the cron_jobs partitioning exercise

- Partition by range - splits one big table into smaller child tables by a value
- Partition key rule - the primary key must include the partition column (unix_timestamp)
- Default partition - safety net child table that catches rows with no matching partition
- Partition pruning - postgres scans only the matching child partitions, verify with explain
- Drop partition - removing old data becomes drop table (instant) instead of delete

UnixTimeStamp:

- it is a reference 10 digit code from the creation of it more like "Since 1970-01-01 00:00:00 UTC that many seconds are completed w.r.t current seconds"
- we can calculate it while remove the current timestamp from the standard timestamp

- the current 10-digit unixtimestamp just shows that many seconds completed since 1970-01-01 00:00:00 UTC